Three Dimensional Geometry

By Naren Ayinala

March 2025

Contents

1 Vectors and Direction Cosines **2**

2 Equation of a Line **2**

3 Angle Between Two Lines **2**

4 Shortest Distance Between Skew Lines **2**

4.1 Parallel Lines 3

5 Finding the Foot of a Perpendicular **3**

6 Image of a Point in a Line **3**

7 Distance from Coordinate Axes **3**

1 Vectors and Direction Cosines

A vector in three dimensions can be written

$$\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}.$$

Its direction ratios are (a, b, c) and magnitude is

$$|\vec{r}| = \sqrt{a^2 + b^2 + c^2}.$$

Its direction cosines are

$$l = \frac{a}{|\vec{r}|}, \quad m = \frac{b}{|\vec{r}|}, \quad n = \frac{c}{|\vec{r}|}, \qquad l^2 + m^2 + n^2 = 1.$$

The direction cosines of the coordinate axes are $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.

2 Equation of a Line

A line through point A with position vector $\vec{a}$ and direction vector $\vec{b}$ has equation

$$\boxed{\vec{r} = \vec{a} + \lambda\vec{b}}.$$

If $A = (x_1, y_1, z_1)$ and $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$, the symmetric form is

$$\boxed{\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}}$$

when a, b, c are nonzero.

3 Angle Between Two Lines

For $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$, the acute angle θ satisfies

$$\cos\theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}.$$

For direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2),

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

Perpendicular lines satisfy $a_1a_2 + b_1b_2 + c_1c_2 = 0$. Parallel lines have proportional direction ratios.

4 Shortest Distance Between Skew Lines

For

$$\vec{r} = \vec{a}_1 + \lambda\vec{b}_1, \qquad \vec{r} = \vec{a}_2 + \mu\vec{b}_2,$$

the shortest distance for nonparallel lines is

$$\boxed{d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}}.$$

For symmetric line equations, substitute the direction ratios and point coordinates into the corresponding scalar triple product.

4.1 Parallel Lines

When $\vec{b}_1 \times \vec{b}_2 = \vec{0}$, the lines are parallel, and

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}_1|}{|\vec{b}_1|}.$$

5 Finding the Foot of a Perpendicular

For a point P and a line $\vec{r} = \vec{a} + \lambda\vec{b}$:

1. Write a general point $Q = \vec{a} + \lambda\vec{b}$ on the line.
2. Use $(\vec{P} - \vec{Q}) \cdot \vec{b} = 0$.
3. Solve for λ and substitute to obtain Q.

The distance is $|\vec{P} - \vec{Q}|$.

6 Image of a Point in a Line

To find the reflection P' of P in a line, first find the foot Q. Since Q is the midpoint,

$$\boxed{\vec{P'} = 2\vec{Q} - \vec{P}}.$$

7 Distance from Coordinate Axes

For a point $P(a, b, c)$,

$$d(P, x \text{ axis}) = \sqrt{b^2 + c^2}$$
$$d(P, y \text{ axis}) = \sqrt{a^2 + c^2}$$
$$d(P, z \text{ axis}) = \sqrt{a^2 + b^2}.$$